

Multiple Choice

1. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Both statements are false because the order of the coset aH does not necessarily divide the order of the element a , and the product of two subgroups HK is not guaranteed to be a subgroup unless one of the subgroups is normal in G .

2. **Answer:** D

Options: A) multiplication is not a binary operation; B) multiplication is not associative; C) identity element does not exist; D) zero has no inverse

Note: The correct answer is based on the fact that for a set to be a group under a binary operation, every element must have an inverse with respect to that operation, and since zero does not have a multiplicative inverse in the set of real numbers, the set fails to satisfy the group axioms.

3. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: The first statement is true because the left cosets and right cosets of a subgroup have the same size, while the second statement is false because left cosets and right cosets may intersect but are not necessarily identical or disjoint.

4. **Answer:** B

Options: A) 0; B) 1; C) 2; D) 3

Note: The correct answer is 1 because in the ring $\mathbb{Z}_3[x]$, the polynomial $x^2 + c$ is irreducible and thus $\mathbb{Z}_3[x]/(x^2 + c)$ is a field if and only if c is not a quadratic residue modulo 3, which occurs only when $c = 1$.

5. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Statement 1 is false because the degree of the sum of two polynomials can be less than the maximum degree of the individual polynomials if they cancel each other out, while Statement 2 is true since in a field, every ideal in the polynomial ring $F[x]$ is generated by a single polynomial, making it a principal ideal.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The ring $2\mathbb{Z}$ has characteristic 0 because there is no positive integer n such that n times any element of $2\mathbb{Z}$ equals 0, indicating that it contains elements of infinite order.

7. **Answer:** False

Options: A) True; B) False

Note: A complete graph with 10 vertices has 45 edges, as the number of edges in a complete graph is given by the formula $n(n-1)/2$, where n is the number of vertices.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because it is a consequence of the formula for the cardinality of the product of two sets, which accounts for the overlap of elements in the intersection of the two subgroups H and K .

9. **Answer:** False

Options: A) True; B) False

Note: The polynomial $x^2 + 1$ evaluates to 1 at both $x = 0$ and $x = 1$ in the finite field $\mathbb{Z}_2$, indicating that it has no zeros in this field.

10. **Answer:** True

Options: A) True; B) False

Note: Every group can be represented as a permutation group by considering the action of the group on itself, thus allowing it to be embedded as a subgroup of the symmetric group on the set of its elements.

Long Form Response

11. **Answer:** To analyze the function $f(x, y) = x^2 - 2xy + y^3$, we begin by finding the critical points by calculating the partial derivatives and setting them to zero. The partial derivatives are $f_x = 2x - 2y$ and $f_y = -2x + 3y^2$. Setting these equal to zero, we find the line $x = y$ as a key constraint, where substituting y for x simplifies our conditions for relative extrema. Specifically, along the line $x = y$, we can rewrite the function as $f(y, y) = -y^2 + y^3$, which allows us to analyze the single-variable function for critical points. Through this analysis, we conclude that all relative extrema of f occur on the line $x = y$, highlighting the significant role this line plays in determining the behavior of the function.

Note: correct because it effectively identifies the critical points of the function by calculating the partial derivatives, demonstrates the significance of the line $x = y$ in simplifying the analysis of relative extrema, and shows how substituting y for x leads to a single-variable function that can be further analyzed for extrema.

12. **Answer:** To analyze the polynomial function $f(x) = 2x^5 + 8x - 7$, we first apply the Intermediate Value Theorem, which suggests that if a continuous function changes signs over an interval, it must have at least one real root within that interval. Evaluating $f(x)$ at several points, we find $f(0) = -7$ and $f(1) = 3$, indicating a sign change between $x = 0$ and $x = 1$, thus confirming at least one real root exists. Furthermore, we can analyze the derivative $f'(x) = 10x^4 + 8$, which is always positive for all real x , indicating that $f(x)$ is strictly increasing. A strictly increasing function can cross the

x-axis only once, leading us to conclude that $f(x)$ has exactly one real root.

Note: correct because it effectively applies the Intermediate Value Theorem to demonstrate a sign change in the polynomial function, indicating the existence of at least one real root between the evaluated points, while also initiating an analysis of the derivative to explore the function's behavior further.

End of Answer Key